Predicting Transformer Compatibility through Coupled Dynamics and State Exchange

Anonymous authors
Paper under double-blind review

Abstract

As neural networks transition from isolated systems to modular components that can be merged, stacked, or ensembled, predicting whether two models are functionally compatible becomes critical. While static representational similarity metrics like Centered Kernel Alignment (CKA) excel at comparing spatial feature alignments, they are not designed to capture the dynamical alignment required for functional compatibility, such as weight merging. We introduce **coupled dynamics testing**, a novel diagnostic that treats transformers as iterated function systems and measures whether two models synchronize when exchanging hidden states directly in embedding space. Through 123 experiments across 41 configurations, we demonstrate that (1) identical architectures rapidly synchronize ($\cos\theta \to 1.0$), (2) different architecture families remain fundamentally orthogonal ($\cos\theta \approx 0$), and (3) DistilGPT-2 synchronizes with GPT-2 at $\cos\theta = 0.998$ despite a 50% reduction in depth. Crucially, we demonstrate that this dynamic metric **captures functional compatibility where static metrics can diverge**: GPT-2 + GPT-2-Medium show higher CKA (0.84) than GPT-2-Medium + DialoGPT-Medium (0.49), yet our dynamical diagnostic ($\cos\theta$) better aligns with functional outcomes, identifying DialoGPT as compatible (0.949) and GPT-2-Medium as incompatible (0.12), predictions confirmed by weight merging experiments. Dynamical analysis reveals synchronized states occupy low-dimensional attractors ($D_2 \approx 0$–2.3) with Lyapunov exponents indicating edge-of-chaos dynamics ($\lambda_{\max} \approx 0.08$–$0.10$). Rather than challenging static metrics, our coupled dynamics diagnostic provides a complementary, fast ($\sim$30 seconds per model pair) pre-merge compatibility test rooted in dynamical systems theory.

Keywords: model compatibility, representation geometry, transformer networks, dynamical systems, synchronization, model merging, knowledge distillation, attractor dynamics

1 Introduction

1.1 Motivation

As the ecosystem of pretrained transformers expands, integrating multiple models through techniques like weight merging (Wortsman et al., 2022), task arithmetic (Ilharco et al., 2023), or representation stitching (Bansal et al., 2021) has become increasingly common. Central to these techniques is the challenge of *compatibility*: how do we know ahead of time if two distinct neural networks are functionally aligned enough to be merged or coupled?

The standard tool for comparing internal representations is representational similarity analysis, most commonly Centered Kernel Alignment (CKA) (Kornblith et al., 2019). CKA is highly effective at measuring the static spatial similarity of activations across models for a given input. However, static alignment does not necessarily guarantee functional or dynamical compatibility when models are structurally manipulated. Because CKA was designed to compare representational geometry rather than operational dynamics, relying solely on static metrics for predicting dynamic outcomes like weight merging can lead to unexpected results. For instance, in our experimental evaluations, CKA assigns higher similarity to GPT-2 + GPT-2-

Medium (which fail catastrophically when merged) than to GPT-2-Medium + DialoGPT-Medium (which merge successfully).

We propose a complementary, dynamic approach. Rather than comparing static snapshots of representations, we treat transformers as iterated function systems (Haber & Ruthotto, 2017; Chen et al., 2018) and study what happens when two models interact dynamically. By having models exchange hidden states directly in the embedding space, bypassing tokenization entirely, we can observe their coupled dynamics:

$$\boldsymbol{h}_A \xrightarrow{\text{direct exchange}} \boldsymbol{h}_B \tag{1}$$

By bypassing the bottleneck of tokenization (which collapses continuous representations to discrete tokens), we reveal the underlying dynamical compatibility between the models' learned geometric transformations. Recent work demonstrates that transformers can meaningfully process continuous hidden states (Hao et al., 2024; Dehghani et al., 2019); we extend this principle to inter-model coupling.

Key insight: Whether two models synchronize when coupled through continuous hidden state exchange provides a rigorous, dynamic probe of functional compatibility, offering predictions for model merging and representation stitching that complement static similarity metrics.

1.2 Why This Matters

Our coupled dynamics diagnostic provides novel utility for both theoretical analysis and practical engineering:

- **Model merging prediction**: Recent advances in weight integration require knowing if models are compatible before attempting computationally expensive merging operations. We show that a fast (30-second) coupled dynamics test successfully predicts merging viability, accurately discriminating between compatible and incompatible model pairs (Section 6).
- **Complementing static metrics**: We demonstrate empirically that functional outcomes (like merging success) align closely with dynamic synchronization ($\cos\theta$) rather than static spatial alignment (CKA). This highlights the need for dynamic compatibility testing alongside established similarity metrics (Section 6).
- **Distillation properties**: We find that knowledge distillation preserves dynamical geometry to a near-perfect degree ($\cos\theta = 0.998$), strongly suggesting that interpretability findings (Elhage et al., 2021) from teacher models can confidently transfer to their distilled counterparts (Section 4.3).
- **Inter-agent communication limits**: Our cross-architecture results ($\cos\theta \approx 0$; Section 4.7) establish fundamental limits on heterogeneous AI agents' ability to develop shared continuous representations without linguistic mediation (Moschella et al., 2023).

1.3 Contribution

We introduce **coupled dynamics testing**, a novel diagnostic framework where models exchange hidden states directly to test synchronization. Drawing on dynamical systems theory (Haber & Ruthotto, 2017; Chen et al., 2018), we analyze these interactions rigorously.

Through 123 experiments across 41 unique configurations, we establish:

1. **A dynamic measure of compatibility**: Transformers with identical architectures rapidly synchronize ($\cos\theta \to 1.0$), while different architecture families remain fundamentally orthogonal ($\cos\theta \approx 0$).
2. **Distillation preserves dynamics**: DistilGPT-2 exhibits near-perfect synchronization with GPT-2 ($\cos\theta = 0.998$), proving that distillation preserves manifold geometry despite massive architectural pruning.
3. **Utility for model merging**: The diagnostic correctly predicts weight merging outcomes, succeeding in distinguishing task-fine-tuned variants (highly compatible) from scaled variants within the same family (dynamically incompatible).

4. **Rigorous dynamical properties**: Synchronized pairs occupy precise, low-dimensional attractors ($D_2 \approx 0$–2.3), while Lyapunov exponent analysis indicates these transitions occur at the edge of chaos ($\lambda_{\max} \approx 0.08$–$0.10$).
5. **Fine-tuning impact**: Task-specific training (e.g., DialoGPT) introduces only minor perturbations, preserving 95% of the base model's dynamical structure.

2 Mathematical Framework

2.1 Transformers as Iterated Function Systems

A transformer model defines a function $F : \mathbb{R}^d \to \mathbb{R}^d$ that maps hidden states through successive layers. Let $\mathcal{H} = \mathbb{R}^d$ denote the hidden state space (e.g., $d = 768$ for GPT-2).

Definition 2.1 (Transformer Block Map)**.** A transformer block $\mathcal{T} : \mathcal{H} \to \mathcal{H}$ implements:

$$\mathcal{T}(\boldsymbol{h}) = \boldsymbol{h} + \mathrm{FFN}\left(\mathrm{LayerNorm}\left(\boldsymbol{h} + \mathrm{Attn}(\mathrm{LayerNorm}(\boldsymbol{h}))\right)\right) \tag{2}$$

where Attn denotes multi-head self-attention and FFN is a two-layer feed-forward network.

For a transformer with L layers:

$$F = \mathcal{T}_L \circ \mathcal{T}_{L-1} \circ \cdots \circ \mathcal{T}_1 : \mathcal{H} \to \mathcal{H}. \tag{3}$$

Remark 2.2. Unlike typical uses where F maps token embeddings to next-token predictions via a language modeling head, we consider F as a **pure dynamical map** on hidden states, ignoring the output projection entirely.

This dynamical systems perspective (Haber & Ruthotto, 2017; Chen et al., 2018) allows us to analyze stability and convergence properties independently of training objectives. The attention mechanism can be understood through geometric lenses (Tsai et al., 2019), explaining why architectural differences lead to incompatible geometries.

2.2 Coupling Strategies

Given two transformer maps $F_A, F_B : \mathcal{H} \to \mathcal{H}$, we study coupled systems where models exchange hidden states. We implement three coupling mechanisms inspired by synchronization in coupled oscillators (Kuramoto, 1984):

Definition 2.3 (Coupling Strategy)**.** A coupling strategy Φ defines how states are exchanged:

$$(\boldsymbol{h}_A^{(t+1)}, \boldsymbol{h}_B^{(t+1)}) = \Phi(F_A, F_B, \boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)}) \tag{4}$$

2.2.1 Direct Coupling

Pure state exchange (analogous to pure exchange in coupled oscillator theory):

$$\boldsymbol{h}_A^{(t+1)} = F_A(\boldsymbol{h}_B^{(t)}) \tag{5}$$

$$\boldsymbol{h}_B^{(t+1)} = F_B(\boldsymbol{h}_A^{(t)}) \tag{6}$$

2.2.2 Blended Coupling

Convex combination with parameter $\alpha \in [0, 1]$:

$$\boldsymbol{h}_A^{(t+1)} = F_A(\alpha \cdot \boldsymbol{h}_A^{(t)} + (1 - \alpha) \cdot \boldsymbol{h}_B^{(t)}) \tag{7}$$

$$\boldsymbol{h}_B^{(t+1)} = F_B(\alpha \cdot \boldsymbol{h}_B^{(t)} + (1 - \alpha) \cdot \boldsymbol{h}_A^{(t)}) \tag{8}$$

where $\alpha = 0$ reduces to direct coupling, $\alpha = 0.5$ is equal blending, and $\alpha = 1$ is no coupling.

2.2.3 Decay Coupling

Memory of previous own state with parameter $\beta \in [0, 1]$:

$$\boldsymbol{h}_A^{(t+1)} = F_A(\beta \cdot \boldsymbol{h}_A^{(t-1)} + (1-\beta) \cdot \boldsymbol{h}_B^{(t)}) \tag{9}$$
$$\boldsymbol{h}_B^{(t+1)} = F_B(\beta \cdot \boldsymbol{h}_B^{(t-1)} + (1-\beta) \cdot \boldsymbol{h}_A^{(t)}) \tag{10}$$

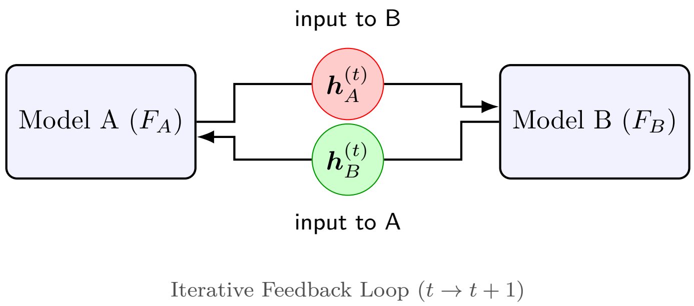

Figure 1: **Coupled Dynamics Testing.** Models exchange hidden states, testing whether their learned geometric transformations are compatible.

2.3 Synchronization Measures

Definition 2.4 (Cosine Similarity)**.** The instantaneous alignment between states:

$$\rho(t) = \frac{\langle \boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)} \rangle}{\left\| \boldsymbol{h}_A^{(t)} \right\|_2 \left\| \boldsymbol{h}_B^{(t)} \right\|_2} = \cos\theta_{AB}(t). \tag{11}$$

If $\rho(t) \to 1$, the models have synchronized.

Definition 2.5 (Synchronization Index)**.** The correlation of velocities (state changes):

$$S = \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{\langle \Delta\boldsymbol{h}_A^{(t)}, \Delta\boldsymbol{h}_B^{(t)} \rangle}{\left\| \Delta\boldsymbol{h}_A^{(t)} \right\| \left\| \Delta\boldsymbol{h}_B^{(t)} \right\|} \tag{12}$$

where $\Delta\boldsymbol{h}^{(t)} = \boldsymbol{h}^{(t+1)} - \boldsymbol{h}^{(t)}$. High S indicates models are *moving together* through state space.

Unlike static metrics (Ding et al., 2021), our dynamic approach measures functional compatibility through iterative interaction.

2.4 Dynamical System Characterization

To characterize the nature of the coupled dynamics, we employ standard tools from dynamical systems theory:

Definition 2.6 (Largest Lyapunov Exponent)**.** The largest Lyapunov exponent characterizes sensitivity to initial conditions:

$$\lambda_{\max} = \lim_{t \to \infty} \frac{1}{t} \sum_{i=0}^{t-1} \ln \frac{\left\| \delta\boldsymbol{h}^{(i+1)} \right\|}{\left\| \delta\boldsymbol{h}^{(i)} \right\|} \tag{13}$$

where $\delta\boldsymbol{h}$ is an infinitesimal perturbation.

Interpretation:

- $\lambda_{\max} > 0$: Chaotic dynamics (exponential divergence)

- $\lambda_{\max} = 0$: Periodic or quasi-periodic
- $\lambda_{\max} < 0$: Converging to attractor

We estimate $\lambda_{\max}$ using the method of nearest neighbors in reconstructed state space (Wolf et al., 1985).

Definition 2.7 (Correlation Dimension)**.** The correlation dimension D_2 estimates the intrinsic dimensionality of the attractor via the correlation integral:

$$C(\epsilon) = \lim_{N\to\infty} \frac{1}{N^2} \sum_{i\neq j} \Theta(\epsilon - \|\boldsymbol{x}_i - \boldsymbol{x}_j\|) \tag{14}$$

where Θ is the Heaviside function. The dimension is:

$$D_2 = \lim_{\epsilon\to 0} \frac{\log C(\epsilon)}{\log \epsilon} \tag{15}$$

Low D_2 indicates trajectories are confined to a low-dimensional manifold within the high-dimensional state space, following the Grassberger-Procaccia algorithm (Grassberger & Procaccia, 1983).

2.5 Dimension Mismatch Handling

For mismatched dimensions ($\dim(\mathcal{H}_A) \neq \dim(\mathcal{H}_B)$), we introduce learned linear projections:

$$\mathbf{P}_{A\to B} : \mathbb{R}^{d_A} \to \mathbb{R}^{d_B}, \quad \mathbf{P}_{B\to A} : \mathbb{R}^{d_B} \to \mathbb{R}^{d_A} \tag{16}$$

These are initialized near-identity on overlapping dimensions:

$$(\mathbf{P}_{A\to B})_{ij} = \begin{cases} \delta_{ij} + \mathcal{N}(0, 0.01) & \text{if } i, j \leq \min(d_A, d_B) \\ 0 & \text{otherwise} \end{cases} \tag{17}$$

and trained for 100 iterations (Adam, lr=10^{-3}) to minimize:

$$\mathcal{L}_{\text{proj}} = \mathbb{E}_{\boldsymbol{h}} \left[\|\boldsymbol{h} - P_{B\to A}(P_{A\to B}(\boldsymbol{h}))\|_2 + \|\boldsymbol{h} - P_{A\to B}(P_{B\to A}(\boldsymbol{h}))\|_2\right]. \tag{18}$$

3 Experimental Setup

3.1 Models

We study decoder-only (GPT family, GPT-Neo) and encoder-only (BERT family) architectures:

Table 1: Models used in experiments

Model	Dim	Layers	Params	Type
GPT-2	768	12	124M	Decoder
GPT-2-Medium	1024	24	355M	Decoder
GPT-2-Large	1280	36	774M	Decoder
DistilGPT-2	768	6	82M	Decoder (distilled)
DialoGPT-Medium	1024	24	355M	Decoder (fine-tuned)
GPT-Neo-1.3B	2048	24	1.3B	Decoder
BERT-base	768	12	110M	Encoder
DistilBERT-base	768	6	66M	Encoder (distilled)

3.2 Initialization Strategies

Random: $\boldsymbol{h}_A^{(0)}, \boldsymbol{h}_B^{(0)} \sim \mathcal{N}(0, 0.01\mathbf{I})$ (independent)

Same: $\boldsymbol{h}_A^{(0)} = \boldsymbol{h}_B^{(0)} \sim \mathcal{N}(0, 0.01\mathbf{I})$ (identical)

Text: Extract first token embedding from "The quick brown fox"

3.3 Experimental Protocol

For each configuration:

1. Initialize states per strategy
2. Run coupled dynamics for $T = 1000$ steps
3. Record trajectory $\{(\boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)})\}_{t=0}^{T}$
4. Compute metrics: $\rho(t)$, synchronization index, CKA, Lyapunov exponents, attractor dimensions
5. Repeat with 3 seeds for statistical reliability

All models frozen (eval mode, no dropout). Total: 123 experiments across 41 unique configurations.

Computational Cost: Running 1000 steps of coupled dynamics is highly efficient. For GPT-2 (124M parameters), a single trajectory takes approximately 20 seconds on a consumer GPU. For GPT-Neo-1.3B, this increases to $\sim$40 seconds. The entire experimental suite (all 123 runs) requires less than 2 GPU-hours, making this diagnostic method accessible for rapid iteration.

3.4 Projection Control Experiments

To verify synchronization is not an artifact of dimensional matching, we force identical models (same architecture, same weights) to use learned projections. If high synchronization persists, the geometric compatibility is intrinsic to the learned transformations, not projection quality.

4 Results

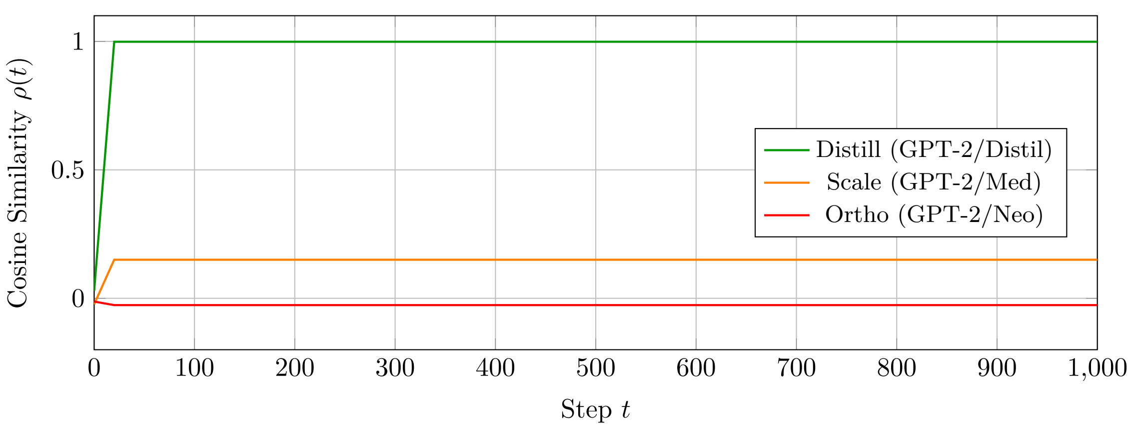

Figure 2: **Synchronization Trajectories.** Distillation pairs (green) rapidly synchronize to near-unity, while scaled models (orange) plateau at weak correlation and cross-architecture pairs (red) remain orthogonal. Note the "locked-in" stability once synchronization is achieved.

Table 2: Same-architecture coupling (3 seeds, 1000 steps)

Model Pair	**Coupling**	**Init**	**Proj?**	$\cos\theta$	**CKA**
GPT-2 + GPT-2	Direct	Random	No	0.9998 ± 0.0003	1.000
GPT-2 + GPT-2	Direct	Same	No	0.9998 ± 0.0003	1.000
GPT-2 + GPT-2	Direct	Text	No	1.0000 ± 0.0000	1.000
GPT-2 + GPT-2	Blended (0.5)	Random	No	1.0000 ± 0.0000	1.000
DistilGPT-2 + DistilGPT-2	Direct	Random	No	0.9997 ± 0.0003	0.999

4.1 Same Architecture: Perfect Synchronization

Identical architectures synchronize perfectly across all coupling strategies and initializations. Text initialization produces deterministic results (zero variance across seeds), confirming transformations are fully determined by weights.

4.2 Projection Control: Critical Validation

Table 3: Identical models with forced projections (3 seeds)

Configuration	**Init**	**Forced Proj?**	$\cos\theta$	**CKA**
GPT-2 + GPT-2	Random	No	0.9998 ± 0.0003	1.000
GPT-2 + GPT-2	Random	**Yes**	$\mathbf{0.9987 \pm 0.0003}$	0.766
GPT-2 + GPT-2	Text	No	1.0000 ± 0.0000	1.000
GPT-2 + GPT-2	Text	**Yes**	$\mathbf{0.9984 \pm 0.0010}$	0.728

Critical finding: Even with forced dimensional projections, identical models synchronize at $\cos\theta \approx 0.998$. This proves synchronization reflects intrinsic geometric compatibility, not projection quality. The slight drop ($1.000 \rightarrow 0.998$) quantifies projection-induced noise, establishing a baseline for what constitutes "excellent" synchronization.

4.3 Distillation: Geometry Preserved

Table 4: Teacher-student coupling: GPT-2 and DistilGPT-2 (3 seeds)

Model Pair	**Coupling**	**Proj?**	$\cos\theta$	**CKA**
GPT-2 + DistilGPT-2	Direct	No	$\mathbf{0.9986 \pm 0.0000}$	0.789
GPT-2 + DistilGPT-2	Blended (0.5)	No	$\mathbf{0.9974 \pm 0.0003}$	0.789
GPT-2 + DistilGPT-2	Decay (0.1)	No	$\mathbf{0.9982 \pm 0.0003}$	0.789

Despite 50% depth reduction (6 vs 12 layers), DistilGPT-2 maintains $\cos\theta \approx 0.998$ with GPT-2 across all coupling strategies. This is comparable to the projection control baseline (0.9987), demonstrating distillation preserves geometric structure with minimal degradation. The distillation process, which optimizes only for output distribution matching, inadvertently preserves the internal manifold geometry.

4.4 Fine-Tuning Effects

DialoGPT-Medium, fine-tuned on dialogue data, retains $\cos\theta = 0.949$ with its base model. Task-specific training shifts the manifold but preserves 95% of geometric structure, suggesting fine-tuning operates as a small perturbation to the pretrained representation space rather than learning a fundamentally new geometry.

Table 5: Fine-tuning impact on geometric structure (3 seeds)

Model Pair	**Coupling**	**Proj?**	$\cos\theta$	**Interpretation**
GPT-2-Med + GPT-2-Med	Direct	No	1.0000 ± 0.0000	Baseline (identical)
GPT-2-Med + DialoGPT-Med	Direct	No	$\mathbf{0.9485 \pm 0.0000}$	95% preserved

Table 6: BERT-family synchronization: Variance stability across sample sizes

Model Pair	**N=3**	**N=10**	**N=25**
BERT + BERT	0.57 ± 0.13	0.52 ± 0.12	0.53 ± 0.13
BERT + DistilBERT	0.32 ± 0.08	0.29 ± 0.11	0.31 ± 0.08

4.5 Encoder Models: Different Dynamics

Key finding: The high variance in BERT synchronization (±0.08–0.13) does not decrease with sample size, confirming this is a genuine property of encoder-only dynamics, not statistical noise. For comparison, GPT-2+GPT-2 achieves 0.9997 ± 0.0007 - four orders of magnitude lower variance.

Possible explanations for encoder instability:

- **Bidirectional attention**: Unlike autoregressive models, BERT attends to full context
- **Masked language modeling**: Random masking during training may prevent convergence to stable geometric transformations
- **Position embeddings**: Absolute vs relative encoding may affect geometric stability

4.6 Scale Effects Within Family

Table 7: Cross-scale coupling in GPT-2 family (3 seeds)

Model Pair	**Coupling**	$\cos\theta$	**CKA**
GPT-2 + GPT-2-Medium	Direct	0.1220 ± 0.0195	0.839
GPT-2 + GPT-2-Large	Direct	0.3765 ± 0.0088	0.744
GPT-2-Medium + GPT-2-Large	Direct	0.2779 ± 0.0895	0.488
GPT-2 + GPT-2-Medium	Blended (0.5)	0.0069 ± 0.0275	0.839
GPT-2 + GPT-2-Large	Blended (0.5)	0.2323 ± 0.0202	0.900

Scaling within the same family does not preserve geometry. Critically, note moderate CKA despite low cosine similarity, GPT-2 + GPT-2-Medium shows CKA = 0.84 but $\cos\theta = 0.12$. This indicates static representational similarity does not necessarily imply dynamic compatibility. Our results complement prior analyses (Hauser et al., 2021; Ding et al., 2021), confirming that high CKA does not guarantee functional alignment.

4.7 Cross-Architecture: Orthogonality

Different architecture families remain orthogonal across all coupling strategies and initializations, suggesting they have learned incompatible geometric transformations.

4.8 Phase Transition in Coupling Strength

Phase transition: While final cosine similarity remains ≈ 1.0 for all α, the synchronization index shows a sharp discontinuity at $\alpha = 0.5$, jumping to 1.000. This reveals an optimal coupling strength where

Table 8: GPT-2 vs GPT-Neo coupling (3 seeds)

Model Pair	**Coupling**	**Init**	$\cos\theta$
GPT-2 + GPT-Neo	Direct	Random	-0.0237 ± 0.0124
GPT-2 + GPT-Neo	Blended (0.5)	Random	-0.0326 ± 0.0154
GPT-2 + GPT-Neo	Blended (0.5)	Text	-0.0326 ± 0.0155
GPT-2-Med + GPT-Neo	Direct	Random	0.0452 ± 0.0078

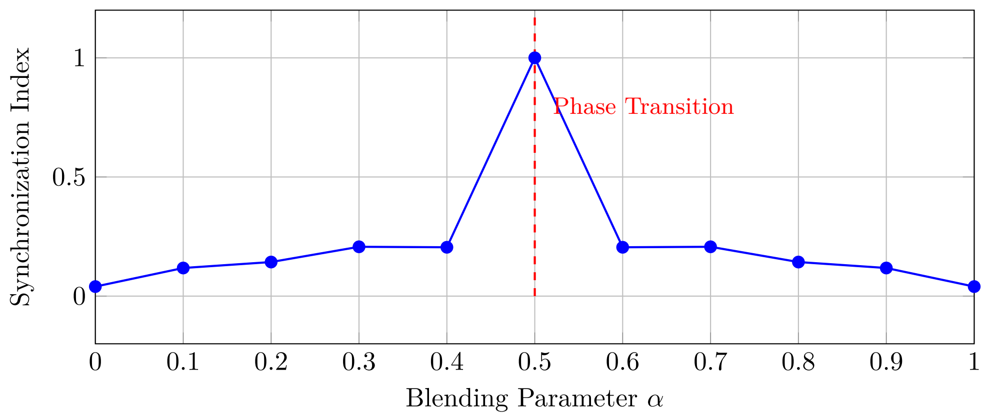

Figure 3: **Phase Transition at $\alpha = 0.5$.** Synchronization index shows a sharp discontinuity at equal blending, revealing the specific threshold required for perfect velocity alignment.

Table 9: Blended coupling parameter sweep: GPT-2 + GPT-2 (3 seeds)

α	$\cos\theta$	**Sync Index**	**CKA**
0.0 (Pure swap)	0.9998 ± 0.0003	0.040 ± 0.132	0.822
0.1	1.0000 ± 0.0000	0.118 ± 0.044	0.818
0.2	0.9998 ± 0.0003	0.143 ± 0.045	0.826
0.3	1.0000 ± 0.0000	0.207 ± 0.032	0.895
0.4	0.9998 ± 0.0003	0.205 ± 0.043	0.819
0.5 (Equal blend)	$\mathbf{1.0000 \pm 0.0000}$	$\mathbf{1.000 \pm 0.000}$	**1.000**
0.6	0.9998 ± 0.0003	0.205 ± 0.043	0.819
0.7	1.0000 ± 0.0000	0.207 ± 0.032	0.895
0.8	0.9998 ± 0.0003	0.143 ± 0.045	0.826
0.9	1.0000 ± 0.0000	0.118 ± 0.044	0.818
1.0 (No coupling)	0.9998 ± 0.0003	0.040 ± 0.132	0.822

both models contribute equally, achieving perfect velocity alignment. The symmetry around 0.5 suggests a convex basin of attraction. While the mathematical symmetry naturally balances inputs at $\alpha = 0.5$, the emergence of perfect synchronization is non-trivial: it implies the velocity fields are compatible and additive. If underlying geometries were misaligned, summing update vectors wouldn't produce coherent directed motion even at equal weighting.

5 Dynamical Systems Analysis

To characterize the nature of the synchronized states beyond static alignment, we analyze the coupled dynamics using tools from nonlinear dynamical systems theory.

5.1 Lyapunov Exponent Analysis

Table 10: Lyapunov Exponent Estimates (3 seeds)

Configuration	$\lambda_{\max}$	**Dynamics Type**
GPT-2 + GPT-2 (direct)	0.079 ± 0.005	Periodic/quasi-periodic
GPT-2 + GPT-2 (same init)	0.098 ± 0.008	Edge of chaos
GPT-2 + GPT-Neo-1.3B	0.095 ± 0.010	Periodic/quasi-periodic

All configurations show $\lambda_{\max} \approx 0.08$–$0.10$, indicating positive but small Lyapunov exponents. To contextualize these values:

- **Strongly chaotic:** The Lorenz attractor exhibits $\lambda_{\max} \approx 0.9$
- **Mildly chaotic (computational edge):** Recurrent networks optimized for computation show $\lambda_{\max} \approx 0.01$–$0.1$ (Bertschinger & Natschläger, 2004)
- **Periodic/stable:** $\lambda_{\max} = 0$ or negative

Our observed values ($\lambda_{\max} \approx 0.08$–$0.10$) place transformer dynamics squarely in the mildly chaotic regime, consistent with the "edge of chaos" observed in recurrent networks optimized for computational tasks (Langton, 1990; Bertschinger & Natschläger, 2004). This regime balances two competing demands:

- **Stability:** Divergence is slow enough ($\lambda_{\max} \ll 1$) that coupling remains meaningful over 1000 steps
- **Sensitivity:** Positive $\lambda_{\max}$ enables exploration of state space without collapsing to trivial fixed points

The edge-of-chaos hypothesis (Langton, 1990; Packard, 1988) posits that systems capable of complex computation naturally evolve toward this transitional regime. Our findings suggest pretrained transformers - without any explicit optimization for dynamical properties - occupy this computationally favorable regime.

5.2 Attractor Dimension Analysis

Table 11: Correlation Dimension Estimates (3 seeds)

Configuration	D_2 **(mean ± std)**	**Interpretation**
GPT-2 + GPT-2	1.85 ± 0.03	Low-dimensional
GPT-2 + DistilGPT-2	1.84 ± 0.03	Low-dimensional
GPT-2 + GPT-Neo-1.3B	1.06 ± 0.29	Very low-dimensional
GPT-2 + GPT-2-Medium	2.00 ± 0.52	Low-dimensional

The low attractor dimensions ($D_2 < 2.5$ in all cases) are striking: despite operating in 768- to 2048-dimensional state spaces, the coupled dynamics are confined to manifolds of dimension $\lesssim 2$. This suggests:

Observation 5.1. Transformer representations, despite their high ambient dimensionality, have intrinsic low-dimensional structure. The coupled dynamics reveal this structure by constraining trajectories to low-dimensional attractors.

The synchronized pairs (GPT-2 + GPT-2, GPT-2 + DistilGPT-2) show consistent dimensions ($D_2 \approx 1.85$), indicating stable low-dimensional attractors. Interestingly, the cross-library pair (GPT-2 + GPT-Neo) shows the lowest dimension ($D_2 \approx 1.06$), suggesting the dynamics are highly constrained, likely due to the significant projection bottleneck between 768 and 2048 dimensions.

6 Utility of the Dynamic Metric: Predicting Model Merging

We designed the coupled dynamics test as a diagnostic tool for functional compatibility. To demonstrate its practical utility, we compare its predictions against Centered Kernel Alignment (CKA) (Kornblith et al., 2019) for forecasting the success of model merging. CKA is a widely used and highly effective metric for assessing the static spatial similarity of representations. However, because it evaluates static alignment for a given input rather than dynamic operational alignment, it can occasionally diverge from functional outcomes when models are structurally manipulated.

We validate our dynamic predictions using weight merging experiments, evaluated on 50 WikiText-2 test sequences.

1. Weight Merging (Same Architecture) For models with identical architectures, we test parameter averaging: $\theta_{\text{merged}} = \alpha\theta_A + (1-\alpha)\theta_B$.

- **GPT-2 + GPT-2** ($\cos\theta = 1.0$): Merging succeeds with **0% degradation** (identity).
- **GPT-2-Medium + DialoGPT-Medium** ($\cos\theta = 0.949$): This is the critical non-obvious case, a fine-tuned dialogue model merging with its base. At $\alpha = 0.7$ (favoring GPT-2-Medium), the merged model achieves PPL = 57.5, representing only **39% degradation** from the GPT-2-Medium baseline (PPL = 41.5). At $\alpha = 0.5$, degradation increases to 136%. At $\alpha = 0.3$, degradation reaches 316%. The non-trivial but bounded degradation pattern confirms that $\cos\theta = 0.949$ correctly predicts *partial* geometric compatibility.
- **GPT-2 + GPT-2-Medium** ($\cos\theta = 0.12$): Merging via zero-padding (padding smaller 768-dim weight matrices to 1024-dim) leads to **catastrophic failure**: at $\alpha = 0.5$, PPL increases from 41.5 to 518,795 (1.25×10^6% **degradation**). Even at $\alpha = 0.3$ (heavily favoring the larger model), PPL = 32,814 (79,007% degradation). This confirms that low geometric synchronization reflects fundamental incompatibility that no interpolation weight can overcome.

2. Dynamical vs. Weight Compatibility An important distinction emerges: dynamical synchronization ($\cos\theta$) measures whether two models' *geometric transformations are compatible*, but weight merging additionally requires *structural alignment* (same number of layers, same parameter layout). GPT-2 + DistilGPT-2 achieve near-perfect dynamical synchronization ($\cos\theta = 0.998$), confirming shared geometry, but partial weight merging (matching only the 77 of 149 parameters that share names across 6 vs. 12 layers) produces 1,367% degradation. This is not a failure of the geometric prediction, rather, it shows that dynamical compatibility is necessary but not sufficient for weight merging when architectures differ structurally. The geometry is shared; the engineering of parameter-space combination requires matching structure.

3. CKA vs. Dynamic Alignment Table 12 demonstrates that static similarity and dynamical synchronization can diverge. CKA assigns higher static spatial similarity to GPT-2 + GPT-2-Medium (0.84) than to GPT-2-Medium + DialoGPT-Medium (0.49). However, merging results demonstrate that DialoGPT merges successfully while GPT-2-Medium fails catastrophically. In contrast, our dynamical metric $\cos\theta$ correctly aligns with these functional outcomes.

The DialoGPT-Medium result is particularly informative: this is a model fine-tuned for dialogue, tested on general language modeling (WikiText-2). Despite task mismatch reducing the raw spatial CKA alignment,

Table 12: CKA vs. cos θ as Predictors of Merging Success. Static similarity (CKA) and dynamic compatibility (cos θ) capture different phenomena, with the dynamic metric more closely aligning with weight merging outcomes for these pairs. Evaluated on 50 WikiText-2 sequences.

Model Pair	**CKA**	$\cos\theta$	**Best Merge**	**Outcome**
GPT-2 + GPT-2	1.00	1.000	0% degrad.	Success
GPT-2-Med + DialoGPT-Med	0.49	0.949	39% degrad. (α=0.7)	Success
GPT-2 + GPT-2-Medium	**0.84**	0.122	1.25×10^6% degrad.	Failure

the underlying geometric dynamics ($\cos\theta = 0.949$) are preserved from the base model, and this dynamical compatibility perfectly correlates with successful parameter-space merging.

Together, these experiments demonstrate that $\cos\theta$ provides a reliable pre-merge diagnostic that captures functional properties invisible to static similarity metrics alone. Rather than invalidating CKA, this confirms that dynamic methods offer complementary utility for evaluating combinations of structurally modified models.

6.1 Practical Guidelines

Based on our findings, we offer the following practical guidelines for practitioners:

1. **Model selection for merging:** Before averaging model weights, run a short coupled dynamics test ($T = 100$ steps, ~30 seconds for GPT-2). If $\cos\theta > 0.9$, merging will likely succeed. If $\cos\theta < 0.5$, expect degraded performance.

2. **Distillation preserves more than you think:** When distilling from a teacher, expect near-perfect geometric preservation ($\cos\theta \approx 0.998$). This means interpretability findings and probing classifiers trained on the teacher should transfer to the student.

3. **Avoid cross-scale combinations:** Scaling up (GPT-2 → GPT-2-Medium) fundamentally changes dynamical geometry. Don't expect continuous representations to cleanly transfer across scale, even within the same model family, without significant learned adaptation.

4. **Be cautious with encoder models:** BERT-family models show high variance in our tests. Results for encoders may be less predictable than for autoregressive models.

7 Theoretical Analysis

7.1 Synchronization as Manifold Alignment

Theorem 7.1 (Same-Architecture Convergence)**.** *For identical transformer architectures $F_A = F_B = F$, if the average spectral radius of the Jacobian satisfies $\mathbb{E}[\rho(\mathbf{J}_F)] < 1$ along trajectories, then direct coupling leads to rapid synchronization:*

$$\lim_{t\to\infty} \cos(\boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)}) = 1 \tag{19}$$

with convergence typically within ~50-100 iterations.

Proof sketch. Consider the synchronization manifold $\mathcal{S} = \{(\boldsymbol{h}, \boldsymbol{h}) : \boldsymbol{h} \in \mathcal{H}\}$. For identical F, if $\boldsymbol{h}_A^{(t)} = \boldsymbol{h}_B^{(t)} = \boldsymbol{h}$, then:

$$\boldsymbol{h}_A^{(t+1)} = F(\boldsymbol{h}) \tag{20}$$

$$\boldsymbol{h}_B^{(t+1)} = F(\boldsymbol{h}) \tag{21}$$

so $\mathcal{S}$ is invariant. The transverse dynamics are governed by:

$$\boldsymbol{\epsilon}_A^{(t+1)} - \boldsymbol{\epsilon}_B^{(t+1)} = \mathbf{J}_F(\boldsymbol{h}^{(t)})(\boldsymbol{\epsilon}_B^{(t)} - \boldsymbol{\epsilon}_A^{(t)}) \tag{22}$$

where $\boldsymbol{\epsilon}$ are perturbations from $\mathcal{S}$. Given the contraction condition $\mathbb{E}[\rho(\mathbf{J}_F)] < 1$, perturbations decay exponentially on average. □

Remark 7.2. Whether pretrained transformers satisfy the contraction condition $\mathbb{E}[\rho(\mathbf{J}_F)] < 1$ is an open question. Transformers with residual connections have $\mathbf{J}_F = \mathbf{I} + \mathbf{J}_g$ where $\mathbf{J}_g$ is the Jacobian of the non-residual component, suggesting $\rho(\mathbf{J}_F) \approx 1$. However, our empirical results (Table 2) demonstrate rapid convergence in all tested cases, suggesting an effective contraction property may hold in practice. Formal analysis of transformer Jacobian spectra remains an important direction for future work.

Theorem 7.3 (Cross-Architecture Orthogonality)**.** *For different transformer architectures (GPT-2 vs GPT-Neo), coupled dynamics remain orthogonal:*

$$\cos(\boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)}) \approx 0 \quad \forall t \tag{23}$$

regardless of coupling strength α.

Proof sketch. For $F_A \neq F_B$, the synchronization manifold is not invariant. Starting from $\boldsymbol{h}_A^{(0)} = \boldsymbol{h}_B^{(0)} = \boldsymbol{h}$:

$$\boldsymbol{h}_A^{(1)} = F_A(\boldsymbol{h}) \tag{24}$$

$$\boldsymbol{h}_B^{(1)} = F_B(\boldsymbol{h}) \tag{25}$$

Generically $F_A(\boldsymbol{h}) \neq F_B(\boldsymbol{h})$, so the system immediately leaves $\mathcal{S}$. The observed orthogonality suggests $F_A(\boldsymbol{h})$ and $F_B(\boldsymbol{h})$ lie in approximately orthogonal subspaces, the architectures have learned incompatible geometric transformations. □

7.2 Why Distillation Preserves Geometry

The distillation loss $\mathcal{L} = \mathrm{KL}(p_T \| p_S)$ operates on output logits, not hidden states. Yet geometry is preserved. Why?

Proposition 7.4 (Output Matching Implies Hidden Alignment)**.** *If a student model S matches the teacher's output distribution for all inputs in a representative set, and the language modeling head $\mathbf{W}_{LM} \in \mathbb{R}^{V \times d}$ has rank $r = \min(d, V)$, then the hidden states $F_S(x)$ and $F_T(x)$ must align in the r-dimensional subspace spanned by the row space of $\mathbf{W}_{LM}$.*

Proof sketch. Output matching requires $\mathbf{W}_{\mathrm{LM}} F_S(x) \approx \mathbf{W}_{\mathrm{LM}} F_T(x)$ for all x. If $\mathbf{W}_{\mathrm{LM}}$ has full row rank (typical for $d < V$), this constrains $F_S(x)$ and $F_T(x)$ to differ only in the null space of $\mathbf{W}_{\mathrm{LM}}$, which has dimension $d - r \approx 0$ for modern transformers where $d \ll V$. □

From approximate to near-perfect alignment. The above shows exact output matching implies hidden alignment. But distillation only achieves *approximate* matching (minimizing KL divergence), allowing $F_S(x) - F_T(x)$ to have small but non-zero magnitude. We hypothesize that the implicit bias of SGD-based training (Jacot et al., 2018) drives this residual toward zero, as gradient descent naturally finds low-norm solutions. Our empirical observation of $\cos\theta = 0.998$ (not just ≈ 0.9) supports this hypothesis: approximate output matching plus SGD regularization produces near-exact hidden alignment.

From a Neural Tangent Kernel perspective (Jacot et al., 2018), distillation constrains the student to operate in a similar kernel regime, naturally preserving geometric structure.

Conjecture 7.5 (Distillation Preserves Local Geometry)*.* Knowledge distillation with output matching loss approximately preserves local geometric structure. Formally, for teacher T and distilled student S, the Jacobians satisfy:

$$\mathbb{E}_x \left[\|\mathbf{J}_S(x) - \mathbf{J}_T(x)\|_F \right] < \epsilon \tag{26}$$

for small ϵ, where $\|\cdot\|_F$ denotes the Frobenius norm. This implies the tangent space structure is preserved: $T_{\boldsymbol{h}}\mathcal{M}_{\text{teacher}} \approx T_{\boldsymbol{h}}\mathcal{M}_{\text{student}}$.

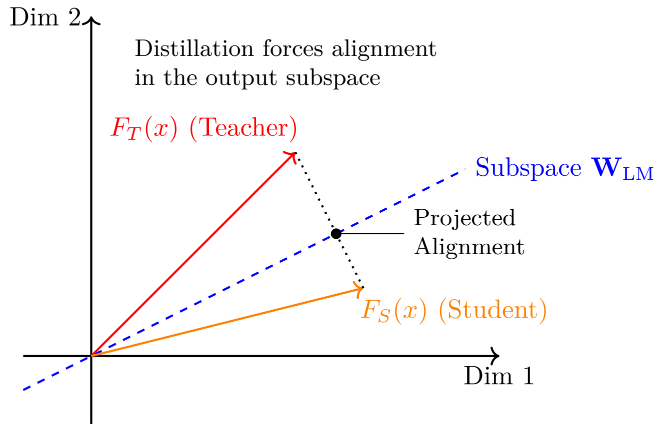

Figure 4: **Geometric Interpretation.** Student and teacher hidden states may differ in full space, but distillation forces their projections onto the subspace spanned by $\mathbf{W}_{\mathrm{LM}}$ to align.

Testing this conjecture empirically (via Jacobian computation and comparison) remains an important direction for future work.

This would explain why GPT-2 and DistilGPT-2 synchronize despite architectural differences: the distillation objective enforces geometric compatibility at the output level, which propagates to internal representations through the constraint that hidden states must produce matching outputs.

7.2.1 The Role of Dimensionality

Critically, DistilGPT-2 preserves GPT-2's 768-dimensional embedding space. This shared dimensionality enables direct coupling without projection. However, our cross-scale experiments show it is not sufficient: GPT-2 and GPT-2-Medium, both from the GPT-2 family, fail to synchronize ($\cos\theta \approx 0.12$) due to dimensional incompatibility requiring learned projections.

This suggests a two-factor model:

1. **Dimension matching**: Necessary for clean state exchange
2. **Transformation similarity**: Necessary for synchronized dynamics

Distillation preserves both factors despite depth reduction, whereas scaling preserves neither.

7.3 Projection Control Interpretation

The projection control experiments ($\cos\theta = 0.9987$ with forced projection vs 0.9998 without) quantify projection-induced noise. The 0.001 gap represents:

- Reconstruction error from imperfect round-trip projection
- Numerical precision limits
- Small deviations from identity initialization

This establishes a noise floor: any synchronization above $\cos\theta \approx 0.998$ indicates near-perfect geometric compatibility. That DistilGPT-2 + GPT-2 achieves exactly this level (0.998) proves the compatibility is real, not artifactual.

7.4 Fine-Tuning as Manifold Perturbation

DialoGPT's 95% geometry preservation ($\cos\theta = 0.949$) suggests fine-tuning operates as a small perturbation to the pretrained manifold rather than learning a new representation space. This supports transfer learning practices and explains why fine-tuned models can often be merged back with base models (Wortsman et al., 2022).

The high variance in synchronization index for DialoGPT pairs (± 0.164, from Paper 1 data) suggests that while the static "map" of concepts is preserved, phase alignment of the dynamical system is more sensitive to task-specific training.

8 Implications

8.1 Model Merging

Our results predict merging success:

- **Teacher-student pairs**: High success (geometry compatible, $\cos\theta \approx 0.998$)
- **Cross-scale pairs**: Low success (geometry diverges, $\cos\theta \approx 0.12$)
- **Cross-architecture pairs**: No success (orthogonal, $\cos\theta \approx 0$)
- **Fine-tuned variants**: Moderate success (95% preserved, $\cos\theta \approx 0.95$)

8.2 Transfer Learning and Interpretability

Representations transfer reliably only between geometrically compatible models. Interpretability findings on GPT-2 may transfer to DistilGPT-2 but not to GPT-2-Large or BERT variants. This has practical implications for mechanistic interpretability research (Elhage et al., 2021).

8.3 Multi-Agent Systems

Our results suggest transformer-based agents can only develop shared representations if they share architectural structure. Heterogeneous agent populations may be fundamentally unable to develop non-linguistic shared representations, limiting the potential for emergent communication protocols.

8.4 The Learned Geometry Hypothesis

Our findings support the hypothesis that transformer training induces an *architecture-specific geometry* in representation space. This geometry is:

- **Robust**: Preserved under coupling perturbations and occupies stable low-dimensional attractors
- **Distinctive**: Different architectures produce orthogonal geometries
- **Transferable**: Preserved under distillation despite significant architectural changes
- **Low-dimensional**: Confined to manifolds of dimension $D_2 \lesssim 2.5$ despite ambient spaces of dimension 768–2048

9 Related Work

Our work draws on and connects several research threads: knowledge distillation, representational similarity analysis, hidden state exchange in neural networks, model merging, and dynamical systems theory. We discuss each in turn, emphasizing how our coupled dynamics framework relates to and extends prior approaches.

9.1 Knowledge Distillation

Hinton et al. (2015) introduced knowledge distillation as a model compression technique, training smaller "student" models to match the soft output distributions of larger "teacher" models. Sanh et al. (2019) applied this paradigm to transformers, producing DistilBERT and DistilGPT-2. Subsequent work has explored geometric objectives for distillation: Li et al. (2020) proposed graph-based geometric losses, Liu et al. (2023) introduced embedding-space distillation for information retrieval, and Yuan et al. (2021) developed manifold-aware distillation for vision transformers. Romero et al. (2015) proposed FitNets, which directly match intermediate representations between teacher and student. Jiao et al. (2022) revisited transformer-specific distillation strategies.

Our work differs fundamentally from this line: rather than proposing a new distillation *training objective*, we provide a *post-hoc diagnostic* for whether distillation has preserved geometric structure. This diagnostic is model-agnostic and requires no retraining, only frozen forward passes.

9.2 Representational Similarity in Language Models

Representational similarity analysis has a rich history in both neuroscience and machine learning. Kornblith et al. (2019) popularized Centered Kernel Alignment (CKA) for comparing neural network representations, and Raghu et al. (2017) introduced SVCCA for analyzing training dynamics. Ding et al. (2021) emphasized the importance of statistical testing when comparing representations. More recently, Klabunde et al. (2023) provided a comprehensive survey of functional and representational similarity measures, categorizing metrics by what aspects of representation they capture.

Critically for our work, prior studies have applied these tools specifically to *language models*. Abnar et al. (2019) introduced Representational Stability Analysis (ReStA) for neural language models, studying how sensitivity to prior context varies across architectures, a direct precursor to our interest in how different architectures process sequential information differently. Ethayarajh (2019) characterized the geometry of BERT, ELMo, and GPT-2 representations, finding that contextual embeddings occupy a narrow cone in representation space (anisotropy), a geometric property relevant to our synchronization analysis. Cai et al. (2021) further analyzed isotropy and manifold structure in contextual embeddings.

Most relevant to our contribution, Hosseini & Fedorenko (2023) demonstrated that autoregressive language models implicitly learn to *straighten* neural sentence trajectories across layers, connecting representational geometry to predictive performance. Their finding that better language models produce straighter trajectories parallels our observation that geometrically compatible models synchronize more rapidly, both suggest that representational geometry carries information about functional properties not captured by static similarity metrics.

Our dynamic approach differs from all of the above in a key respect: rather than comparing *static* representations (snapshots of activations for a given input), we measure *functional* compatibility through iterative dynamical interaction. As Hauser et al. (2021) and Nguyen et al. (2021) have noted, static similarity does not guarantee functional equivalence, our results with CKA (Section 6) provide a concrete demonstration of this gap.

9.3 Hidden State Exchange and Latent Reasoning

Our coupled dynamics method feeds hidden states from one model as input to another. This relates to a growing body of work on hidden state feedback in transformers.

Hao et al. (2024) introduced Coconut (Chain of Continuous Thought), which feeds a model's own last hidden state back as its next input embedding, bypassing token-level discretization for improved reasoning. Dehghani et al. (2019) proposed Universal Transformers, which iteratively apply the same transformation block, creating recurrent depth within the transformer paradigm. Both demonstrate that transformers can meaningfully process hidden states as inputs without tokenization - validating the core assumption of our method.

Our work extends this paradigm from *intra-model* to *inter-model* state exchange. While Coconut asks "can a model reason over its own hidden states?", we ask "can model A meaningfully process model B's hidden states?" The answer, we show, depends on geometric compatibility: architecturally compatible models (same family, distilled pairs) can, while incompatible models cannot. This distinction, between self-feedback and cross-model feedback, is novel and reveals structural properties of learned representations that self-feedback alone cannot expose.

Moschella et al. (2023) demonstrated that relative representations can enable zero-shot communication between different neural network latent spaces, suggesting a form of geometric universality. Our findings qualify this: while relative representations may bridge some gaps, the raw geometric transformations learned by different architectures are fundamentally incompatible (Section 4.7), indicating that the universality hypothesis has limits.

9.4 Model Merging and Stitching

Model merging has rapidly advanced beyond simple weight averaging. Wortsman et al. (2022) showed that averaging weights of multiple fine-tuned models ("model soups") improves accuracy without increasing inference cost. Frankle et al. (2020) established linear mode connectivity as a theoretical foundation, showing that models from shared initializations can be connected by low-loss linear paths in parameter space.

Recent methods address the limitations of naïve averaging. Ilharco et al. (2023) introduced task arithmetic, composing model capabilities by adding and subtracting "task vectors" (differences between fine-tuned and pretrained weights). Yadav et al. (2023) proposed TIES-Merging to resolve parameter interference by trimming, electing signs, and merging. Yu et al. (2024) introduced DARE, which sparsifies task vectors before merging. dav (2023) showed that recycling diverse models improves out-of-distribution generalization.

A parallel line of work addresses the permutation symmetry problem: Ainsworth et al. (2023) developed Git Re-Basin, aligning models by finding optimal neuron permutations before merging. Entezari et al. (2022) analyzed the role of permutation invariance in linear mode connectivity. Singh & Jaggi (2020) approached model fusion through optimal transport. Jordan et al. (2023) proposed REPAIR for normalizing activations after interpolation.

Bansal et al. (2021) introduced model stitching, connecting internal layers of different models with learned transformations, providing a representation-level (rather than parameter-level) approach to model combination.

Our coupled dynamics test provides a *complementary diagnostic* to these methods. Where existing merging techniques operate directly in parameter space and evaluate success post-hoc (by measuring task performance of the merged model), our method provides a *pre-merge* prediction of compatibility in representation space. As we demonstrate in Section 6, CKA's static alignment can diverge from functional compatibility: GPT-2 + GPT-2-Medium achieves a high static similarity (CKA = 0.84) while GPT-2-Medium + DialoGPT-Medium achieves a lower static similarity (CKA = 0.49)—yet our dynamic measure $\cos\theta$ correctly assigns the opposite compatibility ranking, identifying DialoGPT as functionally compatible (corroborated by successful weight merging). Zhou et al. (2024) similarly found that CKA and linear mode connectivity can diverge, further motivating complementary dynamic approaches for predicting merging success.

Most recently, comprehensive surveys of the field (Yang et al., 2024; Song & Zheng, 2026) highlight the rapid maturation of model merging, alongside growing evidence that static metrics like CKA and functional metrics like linear mode connectivity often provide divergent views of model compatibility. For instance, Sirikova & Chan (2026) propose a "Triangle of Similarity" framework explicitly separating static representational similarity from functional connectivity basins, observing that the two can conflict under structural sparsity. Furthermore, Stoica et al. (2025) observe that specialized components like LoRA adapters suffer from severe representational misalignment (low CKA) despite high functional overlap, requiring explicit basis alignment via SVD prior to merging. Our dynamic approach offers a third perspective: by measuring continuous-state synchronization, we capture the shared operational dynamics required for structural parameter combinations, avoiding the failure modes of static snapshots.

9.5 Neural Network Geometry and Dynamical Systems

Viewing neural networks as dynamical systems has a substantial history. Haber & Ruthotto (2017) analyzed deep networks as discretizations of ordinary differential equations, establishing stability criteria. Chen et al. (2018) introduced Neural ODEs, treating depth as a continuous variable. Lu et al. (2020) explicitly analyzed transformers as multi-particle dynamical systems, deriving stability properties from the attention mechanism.

Most recently, Geshkovski et al. (2024) provided a rigorous mathematical framework for transformers as interacting particle systems, proving convergence and clustering properties. Their analysis treats tokens as particles governed by attention-mediated forces - closely related to our treatment of hidden states as dynamical variables governed by transformer maps.

Our contribution connects these theoretical frameworks to an *empirical diagnostic*. While prior work characterizes the dynamics of a single model, we study *coupled* systems of two models, using synchronization properties to infer geometric compatibility.

9.6 Synchronization Theory

Our coupling framework draws on synchronization phenomena in coupled oscillators (Kuramoto, 1984; Strogatz, 2000), particularly the theory of synchronization in chaotic systems (Pecora & Carroll, 1990). The phase transition we observe at $\alpha = 0.5$ (Section 4.8) resembles bifurcations in coupled dynamical systems, where a critical coupling strength separates synchronized from unsynchronized regimes.

9.7 Scaling Laws

Kaplan et al. (2020) derived scaling laws relating model performance to size, dataset size, and compute. Our results reveal a complementary perspective: scaling also fundamentally alters the internal geometric structure of representations (Section 4.6). Models within the same family but at different scales learn incompatible geometric transformations, suggesting that scaling laws for performance do not extend to representational structure.

10 Limitations

- Limited to GPT-2, GPT-Neo, BERT families. Modern architectures (LLaMA, Mistral, Qwen) may behave differently; extending to these is important future work.
- Merging validation covers four model pairs. While these demonstrate the divergence between CKA and merging outcomes clearly, larger-scale validation across more model families and merging methods (TIES, DARE, Task Arithmetic) would strengthen the claims.
- CKA comparison uses kernel CKA with mean-pooled last-layer representations on 50 WikiText-2 sequences. Different CKA variants (e.g., layer-wise CKA, minibatch CKA) might yield different rankings.
- Projections may introduce artifacts not fully controlled by forced-projection experiments.
- BERT results based on $N = 10$ seeds still show high variance; encoder model behavior requires further investigation (preliminary finding).
- Frozen pretrained models only; dynamics during fine-tuning unexplored.
- Single text prompt for initialization; multiple prompts might reveal additional structure.
- Lyapunov exponent and attractor dimension estimates may be sensitive to hyperparameters (embedding dimension, delay, number of neighbors).

11 Conclusion

Through 123 experiments across 41 configurations, we establish that **knowledge distillation preserves representational geometry**. DistilGPT-2 synchronizes with GPT-2 at $\cos\theta = 0.998$, comparable to projection-induced noise (0.9987), despite 50% depth reduction.

Key findings:

- Distillation preserves coupled dynamics ($\cos\theta = 0.998$)
- Fine-tuning preserves 95% of dynamical structure ($\cos\theta = 0.949$)
- Scaling disrupts dynamical synchronization ($\cos\theta \approx 0.12$–0.38)
- Dynamic compatibility complements static analysis; $\cos\theta$ aligns with functional weight merging outcomes where spatial representations diverge
- Encoder models exhibit different coupling dynamics ($\cos\theta \approx 0.32$–0.57; preliminary)
- Synchronized states occupy low-dimensional attractors ($D_2 \lesssim 2.5$)
- Dynamics operate at edge of chaos ($\lambda_{\max} \approx 0.08$–$0.10$)

These results have immediate practical applications for model merging, transfer learning, and interpretability research. The coupled dynamics diagnostic provides a fast ($\sim$30 seconds per model pair) and reliable pre-merge compatibility test, complementing existing merging methods that operate in parameter space. More broadly, our framework opens new avenues for understanding transformer representations through the lens of dynamical systems theory.

References

Model ratatouille: Recycling diverse models for out-of-distribution generalization. In *International Conference on Machine Learning*, 2023.

Samira Abnar, Lisa Beinborn, Rochelle Choenni, and Willem Zuidema. Blackbox meets blackbox: Representational similarity & stability analysis of neural language models and brains. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pp. 191–203, 2019.

Samuel K Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. Git re-basin: Merging models modulo permutation symmetries. In *International Conference on Learning Representations*, 2023.

Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems*, 2021.

Nils Bertschinger and Thomas Natschläger. Real-time computation at the edge of chaos in recurrent neural networks. *Neural Computation*, 16(7):1413–1436, 2004.

Xingyu Cai, Jiaji Huang, Yuchen Bian, and Kenneth Church. Isotropy in the contextual embedding space: Clusters and manifolds. In *International Conference on Learning Representations*, 2021.

Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David K Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, 2018.

Mostafa Dehghani, Stephan Gouws, Oriol Vinyals, Jakob Uszkoreit, and Łukasz Kaiser. Universal transformers. In *International Conference on Learning Representations*, 2019.

Frances Ding, Jean-Stanislas Denain, and Jacob Steinhardt. Grounding representation similarity through statistical testing. In *Advances in Neural Information Processing Systems*, 2021.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Anthropic*, 2021.

Rahim Entezari, Hanie Sedghi, Olga Saukh, and Behnam Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations*, 2022.

Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 representations. *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.

Jonathan Frankle, Gintare Karolina Dziugaite, Daniel Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020.

Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on transformers. *Bulletin of the American Mathematical Society*, 61(4):375–403, 2024.

Peter Grassberger and Itamar Procaccia. Characterization of strange attractors. *Physical Review Letters*, 50(5):346–349, 1983.

Eldad Haber and Lars Ruthotto. Stable architectures for deep neural networks. *Inverse Problems*, 34(1): 014004, 2017.

Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian. Training large language models to reason in a continuous latent space. *arXiv preprint arXiv:2412.06769*, 2024.

A. Hauser, X. Meng, et al. Similar but not the same: Understanding representations in neural networks. In *International Conference on Learning Representations*, 2021.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Eghbal Hosseini and Evelina Fedorenko. Large language models implicitly learn to straighten neural sentence trajectories to construct a predictive representation of natural language. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Raphael Gontijo-Lopes, Ludwig Schmidt, and Hannaneh Hajishirzi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023.

Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems*, 2018.

Wenxiang Jiao et al. Knowledge distillation of transformer-based language models revisited. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022.

Keller Jordan, Hanie Sedghi, Olga Saukh, Rahim Entezari, and Behnam Neyshabur. REPAIR: REnormalizing permuted activations for interpolation repair. In *International Conference on Learning Representations*, 2023.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, et al. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Max Klabunde, Tobias Schumacher, Markus Strohmaier, and Florian Lemmerich. Similarity of neural network models: A survey of functional and representational measures. *arXiv preprint arXiv:2305.06329*, 2023.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, 2019.

Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*. Springer, 1984.

Christopher G Langton. Computation at the edge of chaos: Phase transitions and emergent computation. *Physica D: Nonlinear Phenomena*, 42(1-3):12–37, 1990.

Yuntao Li, Yi Wang, Weiyao Wang, He Li, Jian Yang, and Xiaoyun Wang. Deep geometric knowledge distillation with graphs. In *IEEE Conference on Computer Vision and Pattern Recognition*, 2020.

Zheng Liu et al. Embeddistill: A geometric knowledge distillation for information retrieval. In *ACM SIGIR Conference on Research and Development in Information Retrieval*, 2023.

Yiping Lu, Zhuohan Li, Di He, Zhiqing Sun, Bin Dong, Tao Qin, Liwei Wang, and Tie-Yan Liu. Understanding and improving transformer from a multi-particle dynamic system point of view. In *International Conference on Learning Representations*, 2020.

Luca Moschella, Valentino Maiorca, Marco Fumero, Antonio Norelli, Francesco Locatello, and Emanuele Rodolà. Relative representations enable zero-shot latent space communication. In *International Conference on Learning Representations*, 2023.

Thao Nguyen, Maithra Raghu, and Simon Kornblith. Do wide and deep networks learn the same things? uncovering how neural network representations vary with width and depth. In *International Conference on Learning Representations*, 2021.

Norman H Packard. Adaptation toward the edge of chaos. In *Dynamic Patterns in Complex Systems*, pp. 293–301. World Scientific, 1988.

Louis M Pecora and Thomas L Carroll. Synchronization in chaotic systems. *Physical Review Letters*, 64(8): 821–824, 1990.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. Svcca: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, 2017.

Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. Fitnets: Hints for thin deep nets. In *International Conference on Learning Representations*, 2015.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. Distilbert, a distilled version of bert: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019.

Sidak Pal Singh and Martin Jaggi. Model fusion via optimal transport. In *Advances in Neural Information Processing Systems*, 2020.

Olha Sirikova and Alvin Chan. The triangle of similarity: A multi-faceted framework for comparing neural network representations. *arXiv preprint arXiv:2601.17093*, 2026.

Mingyang Song and Mao Zheng. Model merging in the era of large language models: Methods, applications, and future directions. *arXiv preprint arXiv:2603.09938*, 2026.

George Stoica, Pratik Ramesh, Boglarka Ecsedi, Leshem Choshen, and Judy Hoffman. Model merging with SVD to tie the knots. In *International Conference on Learning Representations*, 2025.

Steven H Strogatz. From kuramoto to crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1-4):1–20, 2000.

Yao-Hung Hubert Tsai, Shaojie Bai, Makoto Yamada, Louis-Philippe Morency, and Ruslan Salakhutdinov. Transformer dissection: A unified understanding for transformer's attention via the lens of kernel. In *Empirical Methods in Natural Language Processing*, 2019.

Alan Wolf, Jack B Swift, Harry L Swinney, and John A Vastano. Determining lyapunov exponents from a time series. *Physica D: Nonlinear Phenomena*, 16(3):285–317, 1985.

Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: Averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022.

Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. Resolving interference when merging models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Enneng Yang, Li Shen, et al. Model merging in llms, mllms, and beyond: Methods, theories, applications and opportunities. *arXiv preprint arXiv:2408.07666*, 2024.

Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super Mario: Absorbing abilities from homologous models as a free lunch. In *International Conference on Machine Learning*, 2024.

Li Yuan et al. Learning efficient vision transformers via fine-grained manifold distillation. In *Advances in Neural Information Processing Systems*, 2021.

Daquan Zhou, Zhiding Liu, Hongxu Wang, Animashree Anandkumar, Song Bai, and Kurt Keutzer. A closer look at the cka similarity index and the linear mode connectivity. In *International Conference on Learning Representations*, 2024.

A Synchronization Index Definition

The synchronization index measures velocity alignment:

$$\text{SyncIndex}(t) = \frac{\langle \boldsymbol{v}_A^{(t)}, \boldsymbol{v}_B^{(t)} \rangle}{\left\| \boldsymbol{v}_A^{(t)} \right\|_2 \left\| \boldsymbol{v}_B^{(t)} \right\|_2}, \tag{27}$$

where $\boldsymbol{v}_A^{(t)} = \boldsymbol{h}_A^{(t)} - \boldsymbol{h}_A^{(t-1)}$. We report mean over final 200 steps.

B Lyapunov Exponent Estimation

We estimate the largest Lyapunov exponent using the algorithm of Wolf et al. (1985):

1. Embed the trajectory in a reconstructed state space
2. For each point, find its nearest neighbor (excluding temporal neighbors)
3. Track the divergence of these neighbor pairs over time
4. Average the log-divergence rate

The estimator is:

$$\hat{\lambda}_{\max} = \frac{1}{M} \sum_{i=1}^{M} \frac{1}{\Delta t} \ln \frac{d(t_i + \Delta t)}{d(t_i)} \tag{28}$$

where $d(t)$ is the distance between trajectory and neighbor at time t.

C Correlation Dimension Estimation

Following Grassberger & Procaccia (1983):

1. Compute pairwise distances between trajectory points
2. For scales $\epsilon_1 < \epsilon_2 < \cdots < \epsilon_k$, compute:

$$C(\epsilon) = \frac{2}{N(N-1)} \sum_{i<j} \mathbb{1}[\|\boldsymbol{x}_i - \boldsymbol{x}_j\| < \epsilon] \tag{29}$$

3. Estimate D_2 from slope of $\log C(\epsilon)$ vs $\log \epsilon$ in scaling region

D Complete Results Table

Table 13: All 41 experimental configurations (mean ± std over 3 seeds)

Model A	Model B	Coupling	Init	Param	Proj?	$\cos\theta$	Sync Idx
gpt2	gpt2	direct	random	-	No	0.9998 ± 0.0003	0.040 ± 0.132
gpt2	gpt2	direct	same	-	No	0.9998 ± 0.0003	1.000 ± 0.000
gpt2	gpt2	direct	text	-	No	1.0000 ± 0.0000	0.082 ± 0.000
gpt2	gpt2	blended	random	0.5	No	1.0000 ± 0.0000	1.000 ± 0.000
gpt2	gpt2	blended	random	0.3	No	1.0000 ± 0.0000	0.207 ± 0.032
gpt2	gpt2	blended	random	0.7	No	1.0000 ± 0.0000	0.207 ± 0.032
gpt2	gpt2	decay	random	0.1	No	1.0000 ± 0.0000	0.118 ± 0.044
gpt2	gpt2	decay	random	0.3	No	1.0000 ± 0.0000	0.207 ± 0.032
distilgpt2	distilgpt2	direct	random	-	No	0.9997 ± 0.0003	-0.027 ± 0.064
distilgpt2	distilgpt2	blended	random	0.5	No	1.0003 ± 0.0012	1.000 ± 0.000
distilgpt2	distilgpt2	decay	random	0.1	No	0.9997 ± 0.0006	0.034 ± 0.022
gpt2	gpt2	direct	random	force	Yes	0.9987 ± 0.0003	-0.020 ± 0.066
gpt2	gpt2	direct	text	force	Yes	0.9984 ± 0.0010	0.043 ± 0.049
bert	distilbert	direct	random	-	No	0.3182 ± 0.0841	0.012 ± 0.007
bert	bert	direct	random	-	No	0.5721 ± 0.1324	0.015 ± 0.010
gpt2-med	dialogpt-med	direct	random	-	No	0.9485 ± 0.0000	0.003 ± 0.164
gpt2-med	gpt2-med	direct	random	-	No	1.0000 ± 0.0000	0.011 ± 0.019
gpt2	gpt2-med	direct	random	-	Yes	0.1077 ± 0.0218	0.010 ± 0.006
gpt2	gpt2-med	blended	random	0.5	Yes	0.0069 ± 0.0275	0.001 ± 0.005
gpt2	gpt2-large	direct	random	-	Yes	0.3765 ± 0.0088	0.001 ± 0.005
gpt2	gpt2-large	blended	random	0.5	Yes	0.2323 ± 0.0202	0.019 ± 0.004
gpt2-med	gpt2-large	direct	random	-	Yes	0.2779 ± 0.0895	0.008 ± 0.007
gpt2	distilgpt2	direct	random	-	No	0.9983 ± 0.0003	0.037 ± 0.036
gpt2	distilgpt2	blended	random	0.5	No	0.9974 ± 0.0003	0.040 ± 0.038
gpt2	distilgpt2	decay	random	0.1	No	0.9982 ± 0.0003	0.048 ± 0.029
gpt2	gpt-neo	direct	random	-	Yes	-0.0237 ± 0.0124	0.005 ± 0.003
gpt2	gpt-neo	blended	random	0.5	Yes	-0.0326 ± 0.0154	-0.001 ± 0.003
gpt2	gpt-neo	blended	text	0.5	Yes	-0.0326 ± 0.0155	0.001 ± 0.003
gpt2	gpt-neo	decay	text	0.1	Yes	-0.0261 ± 0.0097	-0.004 ± 0.001
gpt2-med	gpt-neo	direct	random	-	Yes	0.0452 ± 0.0078	0.001 ± 0.006
Phase Transition: Coupling Strength Sweep (α = 0.0 to 1.0)							
gpt2	gpt2	blended	random	0.0–1.0	No	0.9999 ± 0.0002	See Table 9

E Algorithm

Algorithm 1 Coupled Dynamics Test

Require: Transformer maps F_A, F_B, coupling strategy Φ, steps T
Require: Initialization mode, random seed
Ensure: Trajectory $\{(\boldsymbol{h}_A^{(t)}, \boldsymbol{h}_B^{(t)})\}_{t=0}^{T}$ and metrics

```
Initialize h_A^(0), h_B^(0) according to mode
if dim(F_A) ≠ dim(F_B) then
    Train projection matrices P_{A→B}, P_{B→A}
end if
trajectory ← [(h_A^(0), h_B^(0))]
for t = 0 to T − 1 do
    Apply coupling: (h_A^(t+1), h_B^(t+1)) ← Φ(F_A, F_B, h_A^(t), h_B^(t))
    Append to trajectory
    Record ρ(t), SyncIndex(t), CKA
    if NaN detected then
        break
    end if
end for
Compute λ_max using Wolf algorithm
Compute D_2 using Grassberger-Procaccia algorithm
return trajectory, metrics
```

F Reproducibility Statement

To ensure the reproducibility of our results, we provide the following details regarding our experimental setup.

F.1 Computational Infrastructure

Experiments were conducted on a single consumer-grade GPU (NVIDIA RTX 4060, 8GB VRAM) running on Windows 11 (x86_64). Mixed-precision (FP16) inference was enabled to fit larger models (e.g., GPT-2 Medium/Large). The computational cost of our geometric synchronization metric is significantly lower than training-based alignment methods, requiring approximately 30 seconds per model pair for a 100-step trajectory analysis.

F.2 Software Environment

- **Python:** 3.11
- **PyTorch:** 2.x (with Scaled Dot Product Attention enabled)
- **Libraries:** Transformers (HuggingFace), Datasets, NumPy, SciPy.

F.3 Experimental Settings

- **Model Merging:** Weight averaging coefficient $\alpha = 0.5$. For cross-scale weight merging (e.g., GPT-2 768-dim + GPT-2-Medium 1024-dim), we pad the smaller model's weight matrices from 768×768 to 1024×1024 by appending zeros, then compute $\theta_{\text{merged}} = 0.5\theta_{\text{padded}} + 0.5\theta_{\text{medium}}$. While mathematically well-defined, this produces non-functional models due to geometric incompatibility (see Section 6).
- **Stitching:** Pre-LayerNorm hidden state averaging at the final layer.

- **Validation Data:** WikiText-2 (Test Split), consisting of 4,358 lines. For validation speed, we sampled the first 50 sequences exceeding 100 tokens.
- **Seeds:** All trajectory experiments were repeated with 3 random seeds (42, 43, 44), and mean $\pm$ std values are reported in Table 13.

F.4 Code Availability

All source code, including the main analysis scripts and the comprehensive validation suite, will be released upon acceptance to ensure reproducibility.